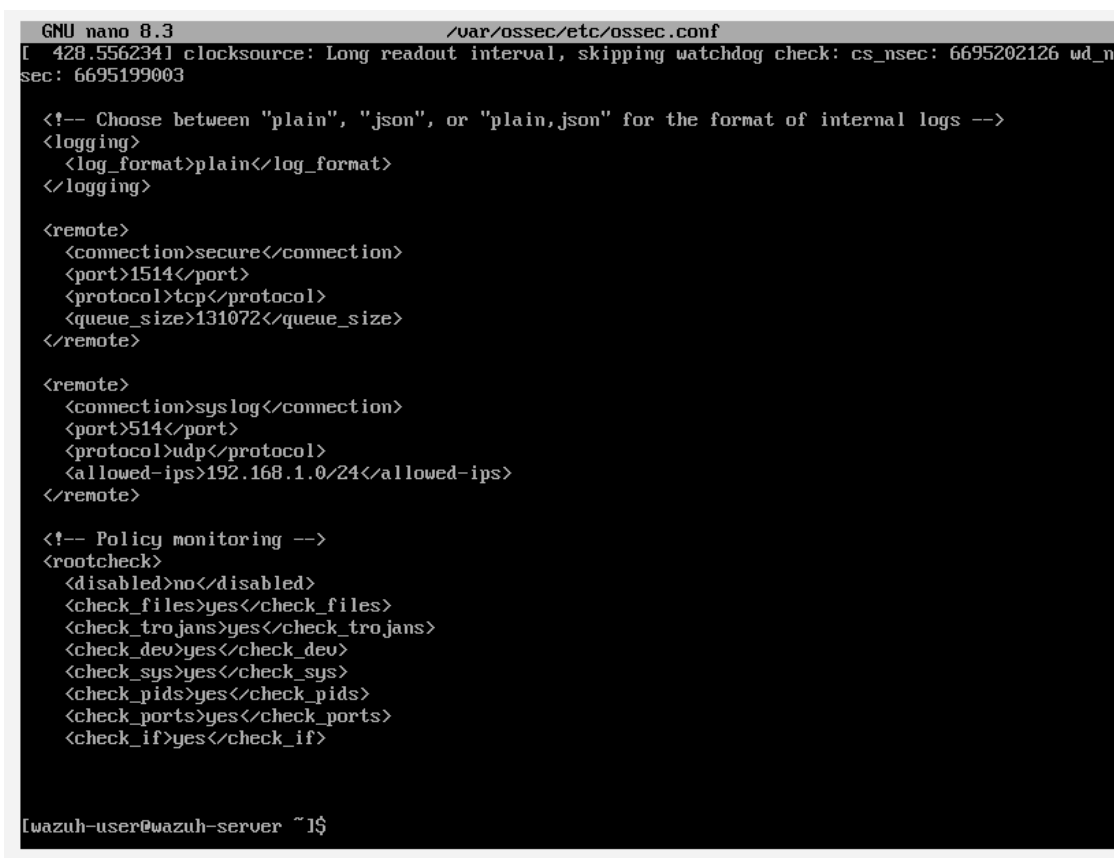


System Configuration: The process begins by configuring the ossec.conf file on the Wazuh manager to define log formats and remote connection parameters, including defining which subnets are allowed to communicate with the server.

Infrastructure Setup: The deployment involves installing and starting the core components of the Elastic Stack specifically Elasticsearch, Logstash, and Kibana on an Ubuntu server using systemctl commands to ensure these services are active and enabled.

Log Collection: Filebeat is installed and configured as the transport agent to forward logs from the Wazuh environment to Elasticsearch. This is verified by checking the Filebeat service status and confirming successful connection attempts to the Elasticsearch URL.

Data Visualization: The final stage requires creating an index pattern (e.g., wazuh-alerts-*) within the Kibana web interface. This allows the platform to ingest, index, and visualize security alert logs, which can then be inspected in the "Discover" tab to confirm that data from the Wazuh server is reaching the dashboard.

A screenshot of a terminal window showing the configuration of the ossec.conf file in nano editor. The terminal title is "GNU nano 8.3 /var/ossec/etc/ossec.conf". The content of the file is as follows:

```
[ 428.556234] clocksource: Long readout interval, skipping watchdog check: cs_nsec: 6695202126 wd_n
sec: 6695199003

<!-- Choose between "plain", "json", or "plain,json" for the format of internal logs -->
<logging>
  <log_format>plain</log_format>
</logging>

<remote>
  <connection>secure</connection>
  <port>1514</port>
  <protocol>tcp</protocol>
  <queue_size>131072</queue_size>
</remote>

<remote>
  <connection>syslog</connection>
  <port>514</port>
  <protocol>udp</protocol>
  <allowed-ips>192.168.1.0/24</allowed-ips>
</remote>

<!-- Policy monitoring -->
<rootcheck>
  <disabled>no</disabled>
  <check_files>yes</check_files>
  <check_trojans>yes</check_trojans>
  <check_dev>yes</check_dev>
  <check_sys>yes</check_sys>
  <check_pids>yes</check_pids>
  <check_ports>yes</check_ports>
  <check_if>yes</check_if>

[wazuh-user@wazuh-server ~]$
```

Figure 1 Configuring Wazuh Manager Configuration File (ossec.conf)

```

Safari/537.36 Edg/92.0.902.67"
2025 Dec 24 07:42:24 wazuh-server->192.168.1.1 Dec 24 07:42:18 nginx: 192.168.1.102 - - [24/Dec/2025
:07:42:18 +0000] "POST /widgets/widgets/interfaces.widget.php HTTP/2.0" 200 855 "https://192.168.1.1
/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/92.0.451
5.131 Safari/537.36 Edg/92.0.902.67"
2025 Dec 24 07:42:25 wazuh-server->192.168.1.1 Dec 24 07:42:19 dhcpcd[871751]: reuse_lease: lease age
18 (secs) under 25% threshold, reply with unaltered, existing lease for 192.168.1.104
2025 Dec 24 07:42:25 wazuh-server->192.168.1.1 Dec 24 07:42:19 dhcpcd[871751]: DHCPREQUEST for 192.168
.1.104 from 08:00:27:67:03:82 (pfSense) via em1
2025 Dec 24 07:42:25 wazuh-server->192.168.1.1 Dec 24 07:42:19 dhcpcd[871751]: DHCPACK on 192.168.1.10
4 to 08:00:27:67:03:82 (pfSense) via em1
2025 Dec 24 07:42:31 wazuh-server->192.168.1.1 Dec 24 07:42:25 nginx: 192.168.1.102 - - [24/Dec/2025
:07:42:25 +0000] "POST /getstats.php HTTP/2.0" 200 125 "https://192.168.1.1/" "Mozilla/5.0 (Windows
NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/92.0.4515.131 Safari/537.36 Edg/9
2.0.902.67"
2025 Dec 24 07:42:36 wazuh-server->192.168.1.1 Dec 24 07:42:30 nginx: 192.168.1.102 - - [24/Dec/2025
:07:42:30 +0000] "POST /widgets/widgets/disks.widget.php HTTP/2.0" 200 545 "https://192.168.1.1/" "M
ozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/92.0.4515.131
Safari/537.36 Edg/92.0.902.67"
2025 Dec 24 07:42:53 wazuh-server->192.168.1.1 Dec 24 07:42:47 nginx: 192.168.1.102 - - [24/Dec/2025
:07:42:47 +0000] "POST /widgets/widgets/interfaces.widget.php HTTP/2.0" 200 855 "https://192.168.1.1
/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/92.0.451
5.131 Safari/537.36 Edg/92.0.902.67"
2025 Dec 24 07:42:55 wazuh-server->192.168.1.1 Dec 24 07:42:49 dhcpcd[871751]: reuse_lease: lease age
48 (secs) under 25% threshold, reply with unaltered, existing lease for 192.168.1.104
2025 Dec 24 07:42:55 wazuh-server->192.168.1.1 Dec 24 07:42:49 dhcpcd[871751]: DHCPREQUEST for 192.168
.1.104 from 08:00:27:67:03:82 (pfSense) via em1
2025 Dec 24 07:42:55 wazuh-server->192.168.1.1 Dec 24 07:42:49 dhcpcd[871751]: DHCPACK on 192.168.1.10
4 to 08:00:27:67:03:82 (pfSense) via em1
2025 Dec 24 07:43:02 wazuh-server->192.168.1.1 Dec 24 07:42:56 filterlog[209471]: 6,,1000000105,em1,
match,block,in,6,0x00,0x33aba,64,UDP,17,36,fe80::a00:27ff:fe67:382,2001:500:2f::f,13422,53,36
2025 Dec 24 07:43:05 wazuh-server->192.168.1.1 Dec 24 07:42:59 ntpd[852581]: error resolving pool 2.p
fsense.pool.ntp.org: Name does not resolve (8)
2025 Dec 24 07:43:06 wazuh-server->192.168.1.1 Dec 24 07:43:00 /usr/sbin/cron[34962]: (root) CMD (/u
sr/sbin/newsyslog)
^C
[wazuh-user@wazuh-server ~]$ █

```

Figure 2 Log from Wazuh Server showing DHCP lease and network activity.

```

Selecting previously unselected package elasticsearch.
(Reading database ... 89632 files and directories currently installed.)
Preparing to unpack .../elasticsearch_7.17.29_amd64.deb ...
Unpacking elasticsearch (7.17.29) ...
Setting up elasticsearch (7.17.29) ...
## NOT starting on installation, please execute the following statements to configure elasticsearch service to start automatically using systemd
sudo systemctl daemon-reload
sudo systemctl enable elasticsearch.service
## You can start elasticsearch service by executing
sudo systemctl start elasticsearch.service
Created elasticsearch keystore in /etc/elasticsearch/elasticsearch.keystore
Scanning processes...
Scanning Linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
evani@ubuntu-server:~$ sudo systemctl start elasticsearch
[sudo] password for evani:

evani@ubuntu-server:~$ sudo systemctl enable elasticsearch
Synchronizing state of elasticsearch.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable elasticsearch
Created symlink /etc/systemd/system/multi-user.target.wants/elasticsearch.service → /usr/lib/systemd/system/elasticsearch.service.
evani@ubuntu-server:~$ sudo systemctl status elasticsearch
● elasticsearch.service - Elasticsearch
   Loaded: loaded (/usr/lib/systemd/system/elasticsearch.service; enabled; preset: enabled)
   Active: active (running) since Sun 2025-12-21 09:46:16 UTC; 58s ago
     Docs: https://www.elastic.co
    Main PID: 3529 (java)
      Tasks: 70 (limit: 4603)
    Memory: 2.2G (peak: 2.2G)
       CPU: 1min 53.080s
    CGroup: /system.slice/elasticsearch.service
            └─3529 /usr/share/elasticsearch/jdk/bin/java -Xshare:auto -Des.networkaddress.cache.ttl=60 -Des.networkaddress.cache.negative.ttl=10 -XX:+AlwaysPre
              └─3710 /usr/share/elasticsearch/modules/x-pack-ml/platform/linux-x86_64/bin/controller

Dec 21 09:45:15 ubuntu-server systemd[1]: Starting elasticsearch.service - Elasticsearch...
Dec 21 09:45:25 ubuntu-server systemd-entrypoint[3529]: Dec 21, 2025 9:45:25 AM sun.util.locale.provider.LocaleProviderAdapter <clinit>
Dec 21 09:45:25 ubuntu-server systemd-entrypoint[3529]: WARNING: COMPAT locale provider will be removed in a future release
Dec 21 09:46:16 ubuntu-server systemd[1]: Started elasticsearch.service - Elasticsearch.
lines 1-16/16 (END)

```

Figure 3 Elasticsearch installation and service startup on Ubuntu server.

```

evani@ubuntu-server:~$ systemctl status logstash
● logstash.service - logstash
   Loaded: loaded (/etc/systemd/system/logstash.service; enabled; preset: enabled)
   Active: active (running) since Thu 2025-12-25 07:23:06 UTC; 8s ago
     Main PID: 1780 (java)
       Tasks: 15 (limit: 4600)
      Memory: 313.4M (peak: 313.8M)
         CPU: 13.871s
    CGroup: /system.slice/logstash.service
            └─1780 /usr/share/logstash/jdk/bin/java -Xms1g -Xmx1g -XX:+UseConcMarkSweepGC -XX:CMSInitiatingOccupancyFraction=75 -XX:+UseCMSInitiatingOccupancy
lines 1-9/9 (END)

```

Figure 4 Logstash service status showing it is active and running on the Ubuntu server

```

Reading state information... Done
The following NEW packages will be installed:
  kibana
0 upgraded, 1 newly installed, 0 to remove and 0 not upgraded.
Need to get 296 MB of archives.
After this operation, 753 MB of additional disk space will be used.
Get:1 https://artifacts.elastic.co/packages/7.x/apt/stable/main amd64 kibana amd64 7.17.29 [296 MB]
Fetched 75.4 MB in 3min 54s (322 kB/s)
Selecting previously unselected package kibana.
(Reading database ... 106356 files and directories currently installed.)
Preparing to unpack .../kibana_7.17.29_amd64.deb ...
Unpacking kibana (7.17.29) ...
Setting up kibana (7.17.29) ...
Creating kibana group... OK
Creating kibana user... OK
Kibana is currently running with legacy OpenSSL providers enabled! For details and instructions on how to disable see https://www.elastic.co/guide/en/kibana/7.1
7/production.html#openssl-legacy-provider
Created Kibana keystore in /etc/kibana/kibana.keystore
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
evani@ubuntu-server:~$ sudo systemctl enable kibana
Synchronizing state of kibana.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable kibana
Created symlink /etc/systemd/system/multi-user.target.wants/kibana.service → /etc/systemd/system/kibana.service.
evani@ubuntu-server:~$ sudo systemctl start kibana
evani@ubuntu-server:~$ sudo systemctl status kibana
● kibana.service - Kibana
   Loaded: loaded (/etc/systemd/system/kibana.service; enabled; preset: enabled)
   Active: active (running) since Sun 2025-12-21 11:19:49 UTC; 10s ago
     Docs: https://www.elastic.co
     Main PID: 7526 (node)
       Tasks: 11 (limit: 4600)
      Memory: 180.5M (peak: 180.5M)
         CPU: 7.469s
    CGroup: /system.slice/kibana.service
            └─7526 /usr/share/kibana/bin/./node/bin/node /usr/share/kibana/bin/./src/cli/dist --logging.dest=/var/log/kibana/kibana.log --pid.file=/run/kibana
Dec 21 11:19:49 ubuntu-server systemd[1]: Started Kibana.service - Kibana.
Dec 21 11:19:49 ubuntu-server kibana[7526]: Kibana is currently running with legacy OpenSSL providers enabled! For details and instructions on how to disable s
lines 1-13/13 (END)

```

Figure 5 Kibana installation and service startup on the Ubuntu server.

```

~
lines 1-10/10 (END)
● filebeat.service - Filebeat sends log files to Logstash or directly to Elasticsearch.
   Loaded: loaded (/usr/lib/systemd/system/filebeat.service; enabled; preset: disabled)
   Active: active (running) since Sat 2026-04-18 02:41:58 UTC; 2 days ago
     Docs: https://www.elastic.co/products/beats/filebeat
    Main PID: 17456 (filebeat)
       Tasks: 7 (limit: 4633)
      Memory: 27.8M
         CPU: 22min 44.279s
    CGroup: /system.slice/filebeat.service
            └─17456 /usr/share/filebeat/bin/filebeat --environment systemd -c /etc/filebeat/filebe
~

```

Figure 6 Filebeat service status confirming it is running.

```
evani@ubuntu-server: ~  
[wazuh-user@wazuh-server ~]$ sudo systemctl start filebeat  
sudo tail -n 50 /var/log/filebeat/filebeat  
2026-03-16T10:53:50.396Z INFO instance/beat.go:645 Home path: [/usr/share/filebeat] Config path: [/etc/filebeat] Data path: [/var/lib/filebeat]  
2026-03-16T10:53:50.398Z INFO instance/beat.go:653 Beat ID: e9a8ffb8-7a61-4f9f-af23-a39214c0519b  
2026-03-16T10:53:50.398Z INFO [index-management] idxmgmt/std.go:184 Set output.elasticsearch.index to 'filebeat-7.10.2' as ILM is enabled  
2026-03-16T10:53:50.399Z INFO eslegclient/connection.go:99 elasticsearch url: http://192.168.1.101:9200  
2026-03-16T10:53:50.501Z INFO [esclientleg] eslegclient/connection.go:314 Attempting to connect to Elasticsearch version 7.17.29  
[wazuh-user@wazuh-server ~]$ cat <<EOF | sudo tee /etc/filebeat/filebeat.yml  
filebeat.modules:  
  - module: wazuh  
    alerts:  
      enabled: true  
  
output.elasticsearch:  
  hosts: ["http://192.168.1.101:9200"]  
  protocol: "http"  
  username: "admin"  
  password: "admin"  
  ssl.enabled: false  
  
setup.ilm.enabled: false  
setup.pack.security.enabled: false  
EOF  
  
sudo systemctl restart filebeat  
filebeat.modules:  
  - module: wazuh  
    alerts:  
      enabled: true  
  
output.elasticsearch:  
  hosts: ["http://192.168.1.101:9200"]  
  protocol: "http"  
  username: "admin"  
  password: "admin"  
  ssl.enabled: false  
  
setup.ilm.enabled: false  
setup.pack.security.enabled: false
```

Figure 7 Filebeat configuration for log forwarding to Elasticsearch.

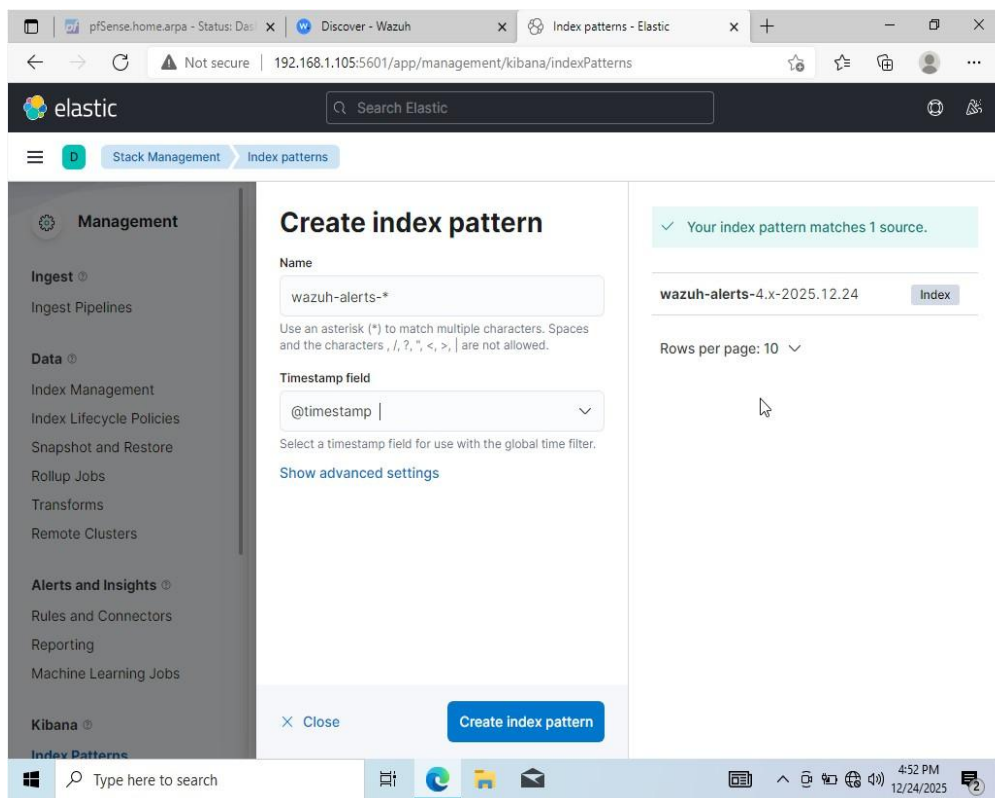


Figure 8 Creating an index pattern for Wazuh alerts in Kibana.

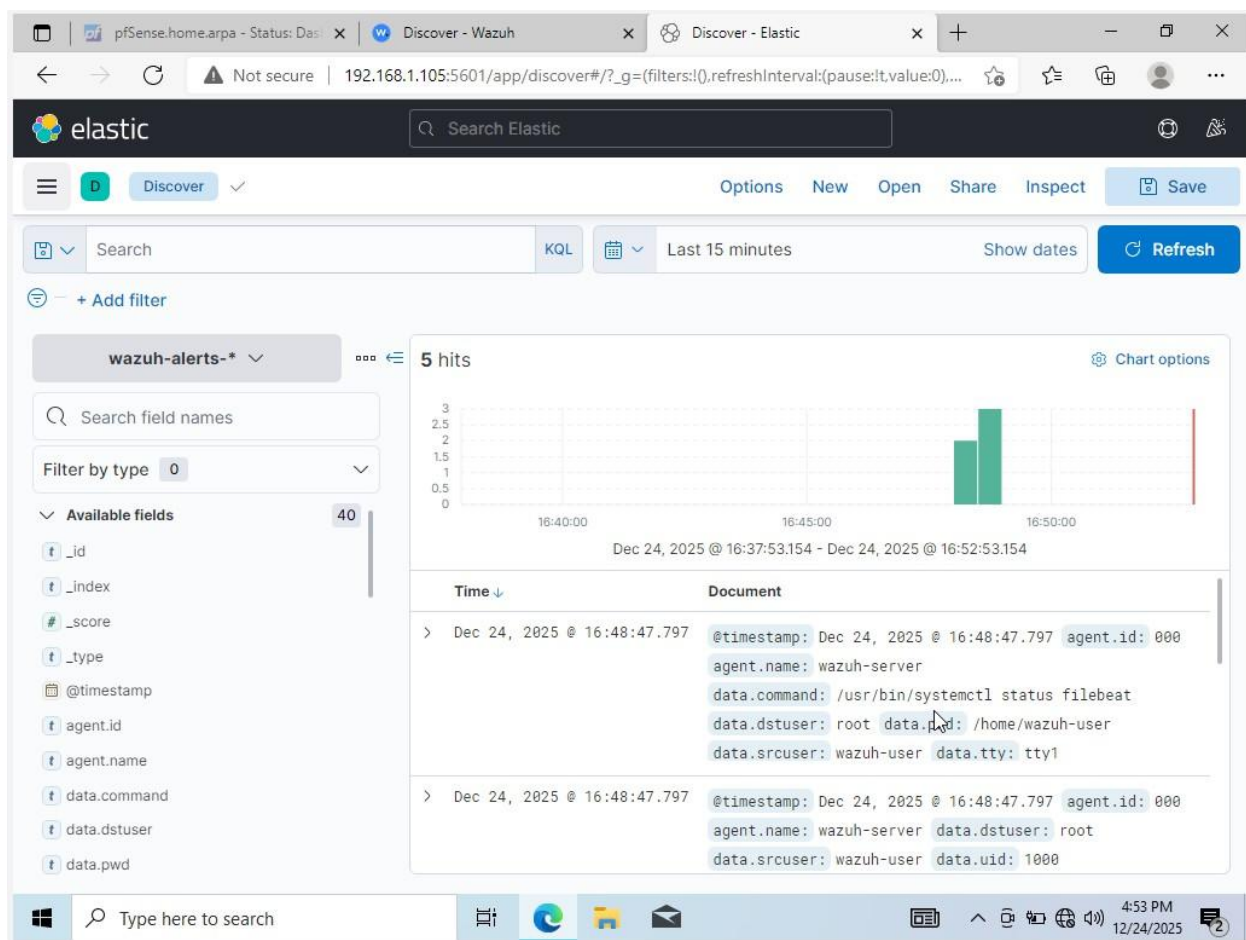


Figure 9 Wazuh alert logs successfully reaching Kibana for visualization

